# Renewable Energy In Bangladesh

**Impact of climate change**

## Pollution Free Real Solution For The Next Generations

**M Golam Mostofa**
Founder & CEO
**Resource Development Foundation**
**RDF**

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(A Prospective Sector)

Pollution Free Real Solution for the Next Generations

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Founder & CEO

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**“To my beloved**
**parents**
**for**
**inspire**
**and**
**guide me**
**To be a philanthropist”**

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However, I started information collection & accumulation of contributory efforts towers the said people by completion of so many Social Development & exclusive development activities along with the Renewable Energy in various areas of the country in 2020.

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M Golam Mostofa

RDF

# Renewable Energy
## (A Prospective Sector)

### Basic Understanding

Renewable Energy is a term "favoring green energy supports that comes only from the natural sources". It is referred to as environment friendly, pollution free & accumulation of smart technologies for sustainable green environmental solutions. Creation of green energy does not impact the earth through toxic green house gases. It is the best solution for lives, livelihoods as well as the clean earth in a greater perspective. Renewable Energy is one of the thrust sectors of the Hon'ble Prime Minister of the Government Republic of Bangladesh. The sector has been growing very fast with its competent, potential, experienced, knowledgeable & committed personnel in its field of operation in Renewable Energy Sector.

The government has setup "committed goal" for providing electricity for all of the inhabitants of the country by uninterrupted supply of green energy at a reasonable and affordable price. Because, sustainable social and economic development depends on adequate power generation capacity of a country. In this aspect there is no other way to accelerate the development except increase in the power generation by fuel diversifications.

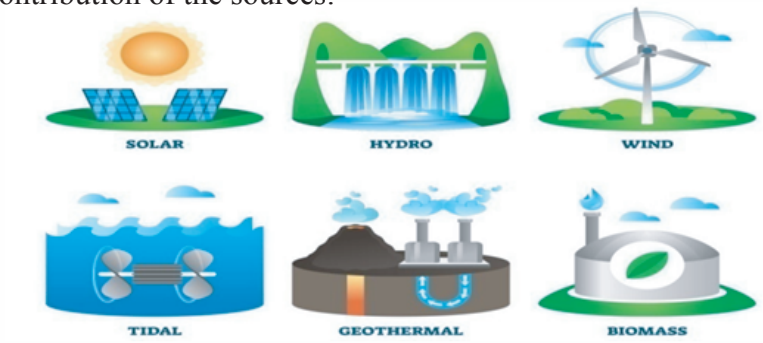
The Sector have been growing very fast by imposing proper knowledge, expertise, skills, potentials as well as commitment of the concern people in Bangladesh; who have in the mean time contributed a lot for its development.

Renewable Energy sources like, Solar System, Hydro Power System, Wind System, Biogas & Biomass System, Geothermal System & Tidal Energy System are the natural sources as well as the contributing avenues of this thrust sector.

Solar radiation depend on the atmospheric image, daytime & season of a year. Bangladesh could apparently meet its unprecedented demand as well as increasing energy supply for energy security by acknowledging the potential of solar-energy resources.

The country is situated between 24o0'0″ N latitude and 90o0'0″ E longitude, which is an ideal position in the earth for preserving & producing solar power. In Bangladesh, the highest level of radiation is accessible from March to April whereas from December to January and the average sunlight hours verification between 6.69/7.6 h, 6.16 h, and 4.81 h respectively in winter, summer and monsoon seasons.

Contribution of the sources:

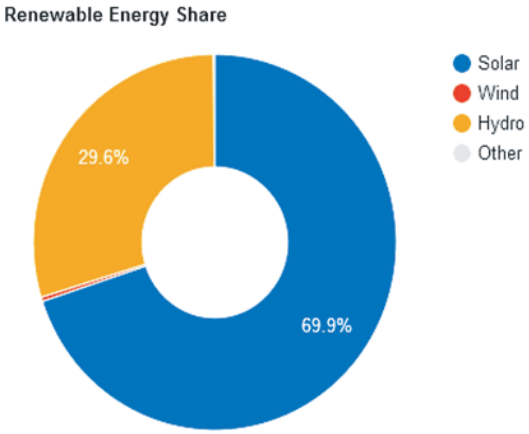



**Energy Contribution in MW**

| Sources | Off-grid | On-grid | Total |
|---|---|---|---|
| Solar System | 347.43 | 195.88 | 543.44 |
| Wind System | 2 | 0.9 | 2.9 |
| Hydro System | 0 | 230 | 230 |
| Biogas & Biomass System | 1.09 | 0 | 1.09 |
| Geothermal System | 0 | 0 | 0 |
| Tidal Energies System | 0 | 0 | 0 |
| **Total** | **350.52** | **426.78** | **777.43** |

Source: Sustainable and Renewable Energy Development Authority (SREDA)

The above data indicates up to 3% energy contribution from renewable energy sector out of the total exist power generation in Bangladesh.




According to the visionary policy & planning of the Government of Bangladesh, followings are the measureable and perfectly applicable issues in the renewable energy sector for accelerating its tremendous growth towards the development of Bangladesh by 2041.

- 8th 5 years plan (2020-2025): It is planned that, total power generation capacity will be achieved up to 30,000MW of which, 10 % will be contributed by renewable energy sources. It indicates that renewable energy will contribute 3000 MW. Out of it, 2014MW will be from solar energy systems and the rest from other avenues.
- SDG-30 (2016-2030): Targeted power generation capacity will be up to 40,000MW of which 4100 MW will be contributed from RE sector & 20% reduction of primary energy consumption by 2030.

- Bangladesh Perspective Plan (2021-2041): Targeted power generation capacity will be 60,000MW aims to reduce production cost, emphasized on Renewable Energy sector through energy efficiency. There is a target of 20% reduction of primary energy consumption by 2031 and 25% by 2041.
- National Solar Energy Action Plan (2021-2041): Power generation capacity is targeted up to 60,000 MW aims to reduce 20% primary energy consumption by 2030 and 25% by 2041.
- The latest addition in the Renewable Energy related policy landscape is the approval of Bangladesh Delta Plan 2100. It is the first time that GOB has ever taken up such 'a long-term strategy to prevent floods and soil erosion, manage rivers and wastes, and supply water throughout the country. The plan is important to the Renewable Energy landscape from multiple viewpoints.
- Mujib Climate Prosperity Plan (MCPP) aims to reduce temperature up to 1.5 degree aligned with Paris Agreement 2015 & produce energy 30% by 2030, 40% by 2041 & 100% by 2050 from Renewable Energy sources. This plan will be achieved within the framework of national planning process (NPP) and will further strengthen the already foreseen the actions of: 8th five years plan, Bangladesh Delta Plan -2100, Vision-2041and Nationally Determined Contributions.
- Our Hon'ble Prime Minister Sheikh Hasina expressed her views in the National Statement in the 26th Session of the United Nations Framework Convention on Climate Change in Glasgow, Scotland on Monday November 1, 2021 that "we submitted an ambitious and updated NDC. Bangladesh has one of the world's most extensive domestic solar energy programs. We hope to have 40% of our energy from renewable sources by 2041. We have cancelled 10 coal-based power plants of worth 12 billion dollars of

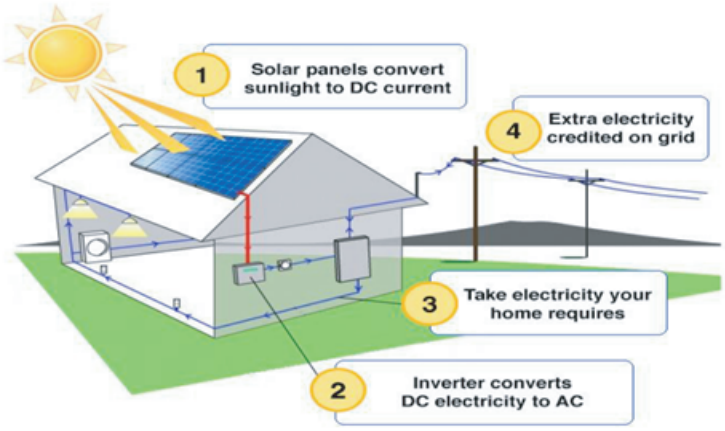
foreign investment. We are going to implement the 'Mujib Climate Prosperity Plan'- a journey from climate vulnerability to resilience to climate prosperity. We are trying to address the challenges of climate impacts because of 1.1 million forcibly displaced Myanmar nationals or Rohingyas".

Solar Systems & its important areas are:

(a) Solar Home System (SHS).
(b) Solar Rooftop System (SRS)
(c) Solar Street Lighting System (SSLS)
(d) Solar Irrigation Pumping System (SIPS)
(e) Solar Net Metering System (SNMS)

**(a) Solar Home System (SHS):**

Technical's of Solar Home System




The Solar System Implementing Agencies in Bangladesh and the partners organization of IDCOL have implemented a big volume of the above mentioned systems.

Increasing access to electricity of rural people through solar home system (SHS) is becoming more popular day by day in Bangladesh. SHS uses PV panels to produce electricity – thereby displacing kerosene conventionally used for lamps as well as diesel used in generators to charge batteries. SHS program was launched to ensure access to clean electricity for the energy starved off-grid rural areas of Bangladesh. The program supplements the Government's vision of ensuring 'SDG-30 & Vision-2041. More than 6 million SHSs have already been installed under IDCOL (the only GoB owned renewable energy platform) program in the off-grid rural areas of Bangladesh and about 18 million beneficiaries are getting solar electricity. The program replaces 1.14 million tons of kerosene having an estimated value of USD 411 million (approx) per year. Moreover, around 75,000 people are directly or indirectly involved with the program. The program has been acclaimed as one of the largest and the fastest growing off-grid renewable energy program in the world.

**How it works?**

Photovoltaic system or solar power system is an electric power system designed to supply usable solar power by means of photovoltaic. It consists of an arrangement of several components, including solar panels to absorb and convert sunlight into electricity, a solar inverter to convert the output from direct to alternating current, as well as mounting, cabling, and other electrical accessories to set up a working system. A solar inverter may connect to a string of solar panels. In some installations a solar micro-inverter is connected at each solar panel. For safety reasons a circuit breaker is provided both on the AC and DC side to enable maintenance. AC output may be connected through an electricity meter into the public grid. The number of modules in the system determines the total DC watts capable of being generated by the solar array; however, the

inverter ultimately governs the amount of AC watts that can be distributed for consumption. For example, a PV system comprising 11 kilowatts DC (kWDC) worth of PV modules, paired with one 10-kilowatt AC (kWAC) inverter, will be limited to the inverter's output of 10 kW. As of 2019, conversion efficiency for state-of-the-art converters reached more than 98 percent. While string inverters are used in residential to medium-sized commercial PV systems, central inverters cover the large commercial and utility-scale market. Market-share for central and string inverters are about 44 percent and 52 percent, respectively, with less than 1 percent for micro-inverters.

Maximum power point tracking (MPPT) is a technique that grid connected inverters use to get the maximum pos.

**(b) Solar Rooftop System (SRS):**

Rooftop solar is a photo voltaic system that has its electricity-generating solar panels mounted on the rooftop of a residential or commercial building or structure. The various components of such a system include photovoltaic modules, mounting systems, cables, solar inverters and other electrical accessories. Rooftop mounted systems are small compared to ground-mounted photo voltaic power stations with capacities in the megawatt range. Rooftop PV systems on residential buildings typically feature a capacity of about 5 to 20 kilowatts (kW), while those mounted on commercial buildings often reach 100 kilowatts even more

Up to July 2021, IDCOL has approved 11Rooftop Solar Projects of which 4 are already in operation with a cumulative capacity of about 3.07 MWp. IDCOL has a target to finance a total capacity of 300 MWp by 2022. industrial and commercial rooftop. The effective utilization of the entire roofs of Bangladesh must ensure excess electricity supports from solar rooftop & may remain a major concern in the country perspective in the energy sector.

IDCOL partner organizations in the meantime have installed a good number of Solar Rooftop solar plants in different areas of the country. RDF has also installed a significant Rooftop Solar Power System supported by IDCOL through QOB funds. RDF is committed to provide contributory supports in Renewable energy sector in Bangladesh.

## Roof top solar PV systems

Conventional power generation involves burning of fossil fuels that releases Greenhouse gas (GHGs) emissions resulting in global warming and climate change. Growing concern about climate change, countries around the world are searching for the best alternative to fossil fuels. Renewable energy is the solution but current state of technology and cost factor makes it the least suitable option as an alternate choice. However, since the past one decade, the technology has evolved in such a way that we can see utility scale renewable energy power projects based on Solar, Wind and other renewable technologies coming up. For countries like India where coal is the dominant fossil fuel to generate electricity, the least GHG emission intensive option is the major requirement. Rooftop Solar PV system is the best option nowadays.

Unlike MW scale solar PV projects, rooftop system uses very less space and is also cost effective. We know that the major issue behind utility scale solar PV project is availability of land. It takes around 3 acres of land for installing 1MW solar PV plant. In Bangladesh, where the population is growing enormously, we require considerable land area for growing food grains to support the growing population. Agricultural land conversion for solar project does not seem a viable option. Another issue is the availability of grid. Rural population is still struggling for electricity as they don't have access to power grid. In addition, there is acute shortage of electricity generation due to non/ inadequate availability of coal resulting in frequent power cuts.

## How it works?

Solar PV rooftop system is basically a small power plant at our rooftop. The Grid interactive Roof Top Solar Photo Voltaic (PV) mainly consists of three major components. These are the solar PV modules, mounting structure for the modules and the inverter or power conditioning units. Solar PV modules form an array and it requires a mounting structure to hold PV modules at the required angle for maximized generation.

The solar panels convert solar energy in the form of light into electricity in DC form (Direct Current). The DC electrical energy is converted to AC (Alternate Current) power by the inverter/power conditioning unit which is connected to the power grid through AC distribution board. The AC power output can be measured through metering panel connected to it. The 415 V AC output of the system can be synchronized with the grid and the electricity can be exported to the grid depending upon solar power generation and local consumption.

**Grid synchronization**

Electrical inverters convert direct current generated from solar PV modules to alternating current. Therefore, solar modules need to be connected to inverters. Nowadays we have smart Grid-interactive inverters. These inverters can produce AC power that matches voltage and frequency of the grid and the power line it connects to. There is another requirement of adding an isolation transformer. For safety purposes, isolation transformer is used to transfer electrical power from inverter to the connected load while isolating the load from the power source. In addition, the injection of DC power into the grid can be avoided by using an isolation transformer at the output of the inverter.

Power generated from the rooftop solar system during the daytime can be utilized fully by powering the building loads and feeding excess power to the grid as long as grid is available. Whenever, solar power is not available due to shadow or a cloudy day, the building loads can be served by drawing power from the grid or DG sets. Smart inverters automatically understand the power situation and always give preference to power generated from solar modules.

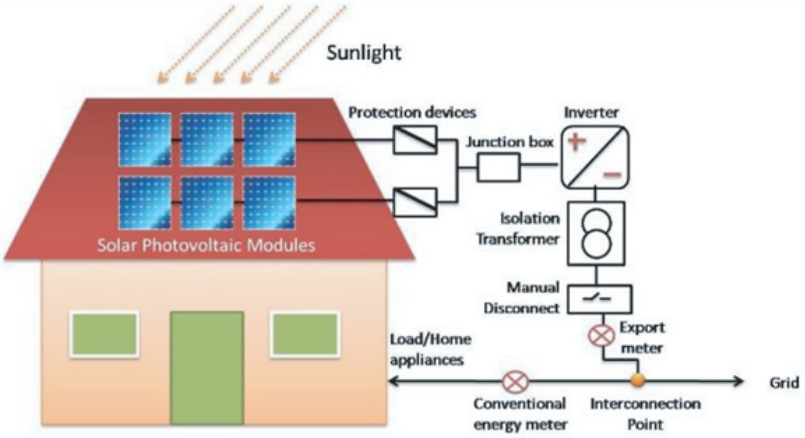



Fig 1: Feed in metering schematic

**Key factors of rooftop solar systems;**

- The solar panels should be south facing to maximize the production (In Bangladesh). The angle of panel should be 25-30 degrees with respect to the land surface
- The weight exerted by the Solar Panels and the structure is around 10 Kgs per square meter.
- The solar installation require about 100-150 square feet of area per Kilowatt.
- No frequent maintenance is required for the solar system except regular cleaning of surface of the panels.
- There are two types of rooftop solar installations. First is for captive use and second is for feed in tariff based grid connected system. In captive use, rooftop solar system installer himself consumes all electricity generated by the system. However, in second type i.e. feed in tariff based grid connected system, installer can also feed/sale excess power to the grid.
- In captive rooftop solar systems, there are two types. The first one is standalone systems and other is grid connected system.

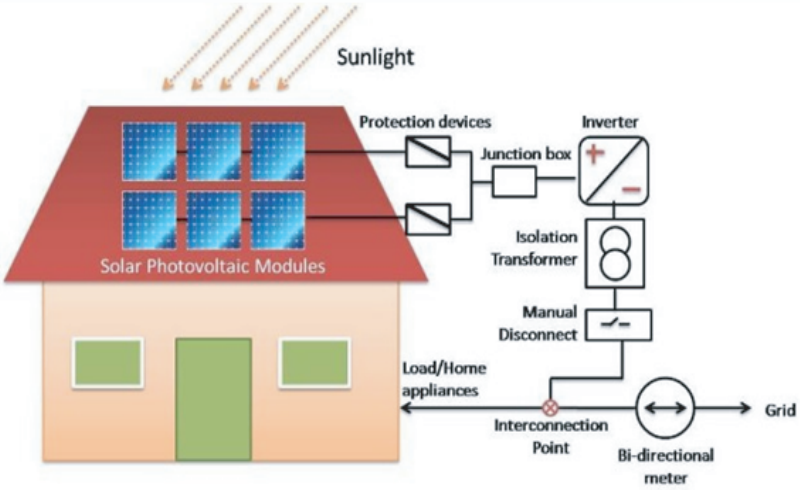
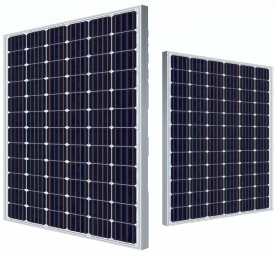


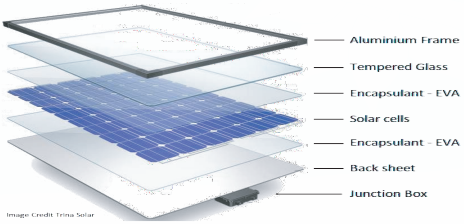

Fig 2: Net metering schematic

## Benefits of Rooftop Solar PV systems

- Cost of electricity is increasing. Electricity utility companies are bound to increase the cost per unit of electricity due to increase in cost of fuel, Many regions in the country are facing severe power cuts extending to 12 -14 hours in a day, Grid tied solar PV systems are the most common systems for captive power generation. It is technologically feasible to have hybrid systems such as Solar PV +Grid or Solar PV +Diesel Generator, Ensuring Uninterrupted Power supply, Cheaper electricity supports to the national grid, Utilization of unused spaces, Easy electricity to the national grid, Reduced use of grid electricity, Full utilization of national spars, Easy implementation & hassle free operation, Pollution free Environment, Mitigate own consumption of required electricity.

## Solar PV System Components






### Solar Modules

- Poly Crystalline
- Mono Crystalline [PERC, Efficiency 20%]

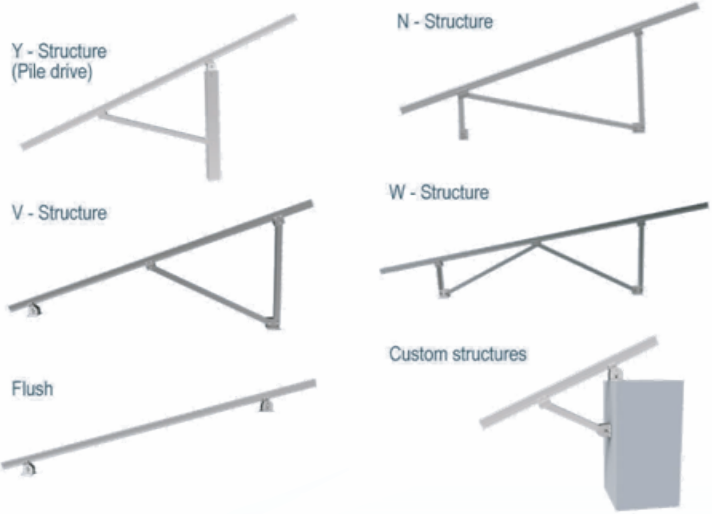
### String Inverter

5kW ~150kW

Pros: Fan Less, low maintenance

Cons: Higher BOS cost

Mounting Structures



**Types:**

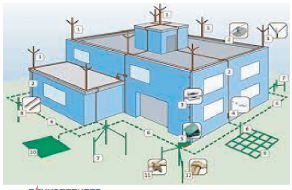
- MS
- Hot dip Galvanized
- Anodized Aluminum (Light Weight)

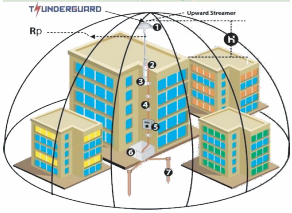
**Balance of System (BOS)**

UV resistant
DC cables (PV1-F)

Copper

Early Streamer






**(c) Solar Street Lighting System (SSLS):**

Solar LED Street Lighting now a day is the real demand from the Rural People of Bangladesh. It makes sustainable advantages of new clean energy technologies in the form of smart solar LED lights, while saving sufficient amount of expenses in the process.

In the result past, most of the traditional street lighting was sodium system. They are polluting the environment. Massive efforts are being made world-wide to replace traditional street lights. At this

moment there are huge odifications taking places, some advantages of solar LED lights compared to traditional, lights that may raise the eyebrows of parking lots owners and municipalities.

Solar streets lighting system now a days became popular in GoB policy and decision makers. It is now secured by lithium iron phosphate batteries fixed in the body of lighting system.

Lastly, solar street lighting ensures security of the society & provides supports to smart livelihood of the rural as well as urban people of Bangladesh.

Why Solar Street Light?

- Sustainable Energy Efficiency;
- Smart Technology & automatically operation;
- Luminosity & economy of voltage consumption;
- Pollution free environment;
- Ensuring Security & shining the villages;

(d) Solar Irrigation Pumping System (SIPS)

It's the ground reality that fuel & electricity based irrigation system is still predominant in Bangladesh and farmers rely on it for their irrigation management. About 1.34 million diesel pumps and 0.27 million electric pumps are operating across the country in agricultural sector for irrigation. As per World Bank reports, 1 million MT diesel of worthing US$ 900 million are being consumed every year for irrigation. Obviously, this is a massive demand for fuel which GOB subsidizes at the opportunity cost of other agricultural innovation. Furthermore, diesel based irrigation system is not environment friendly as significant amount of green house gases

is being emitted to pollute the environment as well as demandable electricity bound the government to expend big amount of foreign currency per year; which directly affected the economy through rental electricity commissioning purchase. In order to drive agriculture sector and to provide access to continue energy supply & supports to the farmers, diesel & electricity based irrigation system needs to be converted by solar irrigation pumping system (SIPS). Solar energy powered pumps are reliable as irrigation alternatives for farmers where solar technology helps the country in various ways which are in the process of though management of the related concern.

**Its impacts & supports:**

Multiple impacts in agricultural sector has been created by this SIPS System like pollution free environment, hassle free supports to the farmers, reduces dependency on electricity & diesel addressing nothing botheration, 43% irrigation cost reduction, Positive impacts on climate change through reducing Carbon emission as well as save millions of foreign currency in every year.

It is solar PV-Module based irrigation systems creating natural electricity by sun ray in cultivating areas where there are possibilities to produce three types of crops throughout the year.

Practically, it should be in the thought process that each solar pump replaces 10 to 12 diesel pumps and covers more than 150 bighas of land producing 3 (three) cycle crops in each year. On the basis of life cycle costs, solar irrigation pumps are much more attractive cost economy than compared to diesel & electricity driven pumps. In addition, solar PV pumping systems allow low operating cost, unattended operation, low maintenance, easy installation and long lasting life cycle devices.

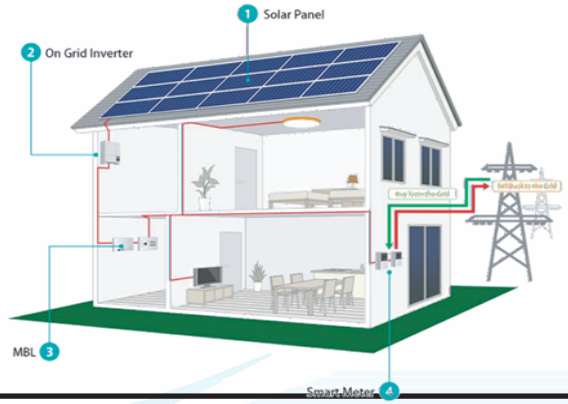
Solar irrigation pumping system (SIPS) is innovative and environment friendly solution for agro-based economy of Bangladesh. This system reduces dependency on fuel and demand for electricity from national grid in irrigation seasons. The program also reduces carbon emission and at the same time saves millions of foreign currency.

**GoB initiatives:**

The Hon'ble Prime Minister Sheikh Hasina has taken initiative to accelerate the activities for growing more food as well as reducing consumption of grid electricity for reducing Carbon emission. IDCOL, the government owned organization has been providing supports to its implementing organizations for fulfillment of Hon'ble Prime Minister's 30% energy in "SDG-30" through Green Energy supports and SIPS is one of them.

Meanwhile, IDCOL has approved 1,515 solar irrigation pumps are operational with an installed capacity of around 40 MW. The World Bank, KfW, GPOBA, JICA, USAID, ADB and BCCRF are providing financial supports for those initiatives.

The Government has set up a target to install/replace the diesel & electricity operated pumps by a significant solar pumps which will ensure 150 MW energy generations from SIPS.



**Solar Net Metering System (SNMS):**

The Government of Bangladesh has recently developed the power generation strategy based on fuel diversification to enhance energy security. Alongside the conventional energy sources, renewable energy will play a significant role in meeting the future demand of electricity as well as fulfilling environmental obligations. Bangladesh adopted, keeping pace with the global trend, The Renewable Energy Policy of Bangladesh mandates that 40 % of electricity to come from renewable energy sources by 2041.

Most of the recent capacity addition is from standalone solar systems, generally referred to as solar home systems (SHSs). There are 6 million cumulative installations of SHSs in predominantly off-grid areas of Bangladesh. Making it the largest off grid renewable energy program in the world. Despite huge potential, grid-connected electricity consumers are yet to reap the benefits of solar energy. Every on-grid household and commercial or industrial consumers can utilize solar energy, which is the most dispersed form of energy, to generate electricity by installing solar photovoltaic (PV) panels on their own roofs and can become electricity producers meeting their electricity demand partly or fully by themselves and can even sell excess electricity produced to the distribution utilities if appropriate policies are in place.

Bangladesh enjoys good amount of sunshine and the use of solar energy continues to grow while the cost of solar technology continues to decline. Incentivizing grid-connected customers is of utmost importance to promote RE-based distributed generation. Net metering is one of the tools to popularize the RE based electricity generation in the country. Net metering is a policy approach designed to encourage distributed renewable energy development by allowing utility customers to generate their own electricity from solar or any other renewable sources and use the

electricity produced to offset the amount of energy they draw from the utility grid (sometimes called the distribution grid) and any access generation can feed into the grid. Customers are only billed for their "net" energy use and receive credit usually in the form of kilowatt-hour (kWh) during a given period. A net balance in favor of the customer is carried forward to the next month, while a balance in favor of the utility is settled at the end of the month as usual.

Net-metering can potentially drive widespread implementation of distributed generation by incentivizing end-users to adopt localized power generation through renewable energy technologies (RETs) such as solar, wind and biomass. As of 2017, 46 countries have some form of active net metering policy; local governments have adopted net metering policies in another nine countries in the absence of national-level actions.

Realizing its importance, the Government of Bangladesh published a net metering guideline in July, 2018 to establish a mechanism for distributed RE integration to the grid.

**RDF Tower**

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